

① Circular Motion ^{*}Fundamentals:

Uniform C.M.

$$(\omega = \text{const})$$

$$\alpha = 0$$

Non uniform C.M.

$$\omega \neq \text{Constant}$$

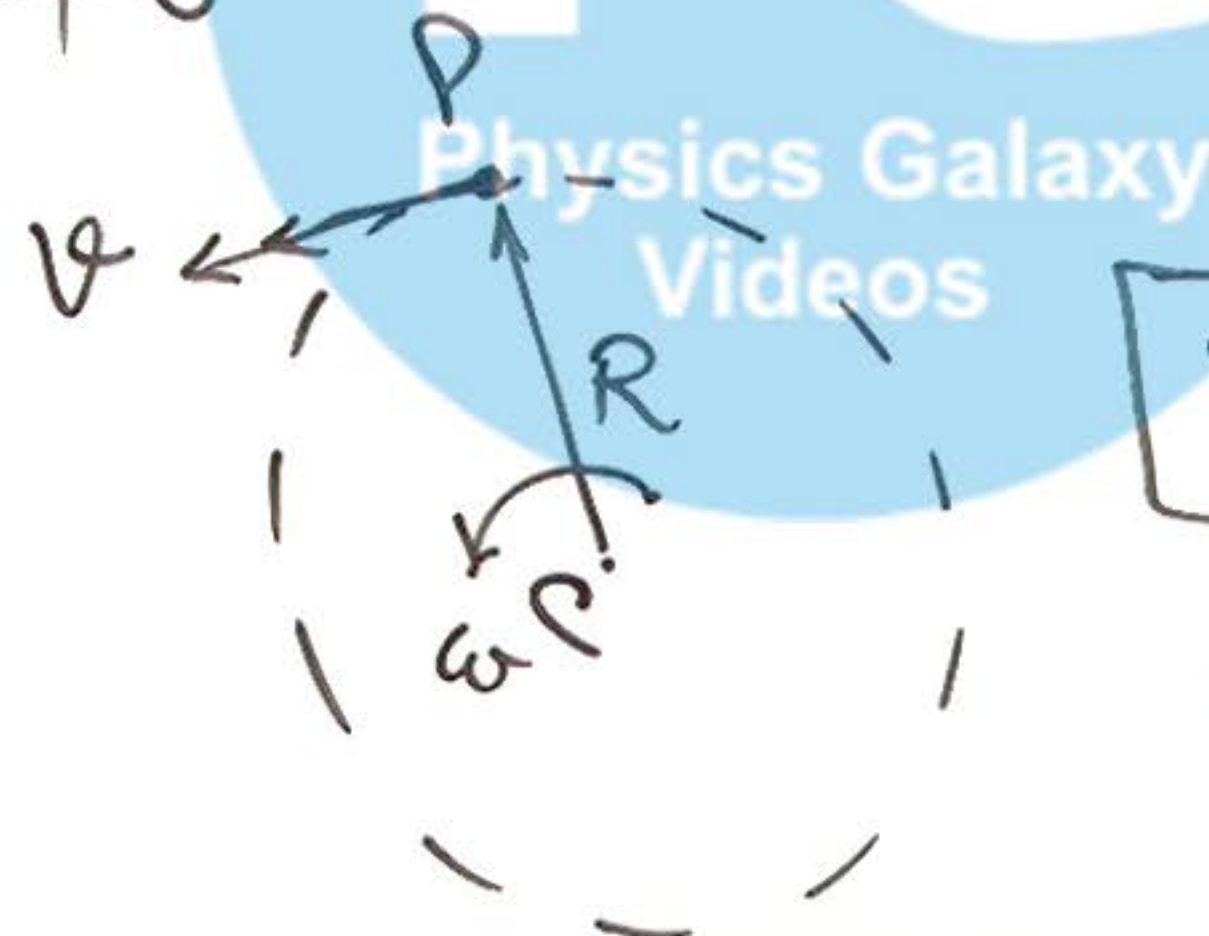
$$\alpha \neq 0$$

→ for a point mass.

- Angular Displacement (θ or $\Delta\theta$) rad
- Angular Velocity (ω) rad/s
- Angular Acceleration (α) rad/s²

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt} = \omega \frac{d\omega}{d\theta}$$



$$v = R\omega$$

$$\frac{dv}{dt} = R \frac{d\omega}{dt}$$

tangential accⁿ

$$a_t = R\alpha$$

② Tangential & Normal Acceleration in 2D:

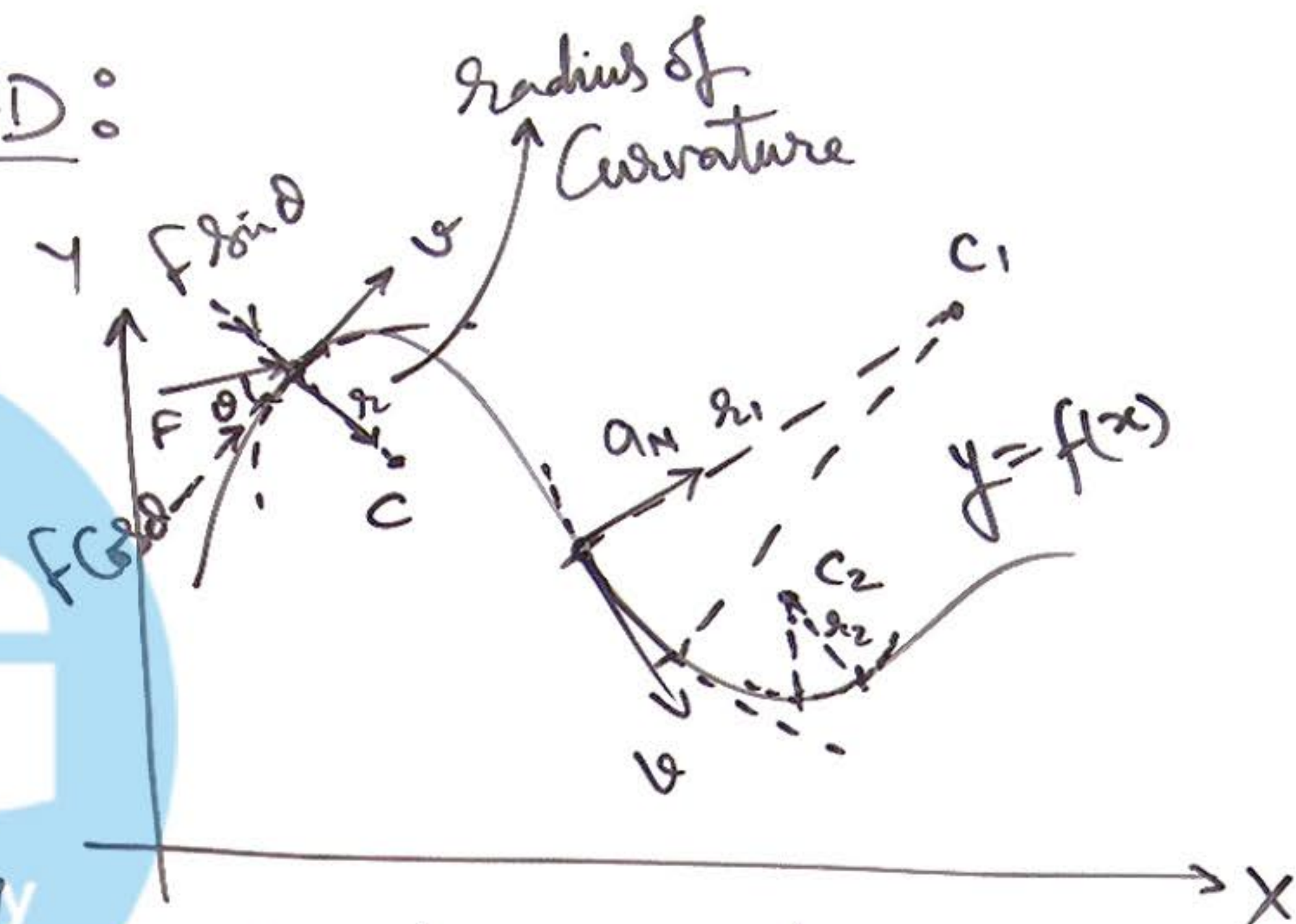
$$a_t = \frac{F \cos \theta}{m} = \frac{dv}{dt} \quad a_N = \frac{F \sin \theta}{m} = \frac{v^2}{r}$$

Total accⁿ $a_T = \sqrt{a_t^2 + a_N^2}$

for a given eqⁿ of traj $y = f(x)$

Rad of curvature

$$r = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\frac{d^2y}{dx^2}\right|} \quad (x, y)$$



Curvature $\rho \propto \frac{1}{\text{Rad of curvature } (r)}$
 ↓
 Bend in Curve

③ Uniform Circular Motion :

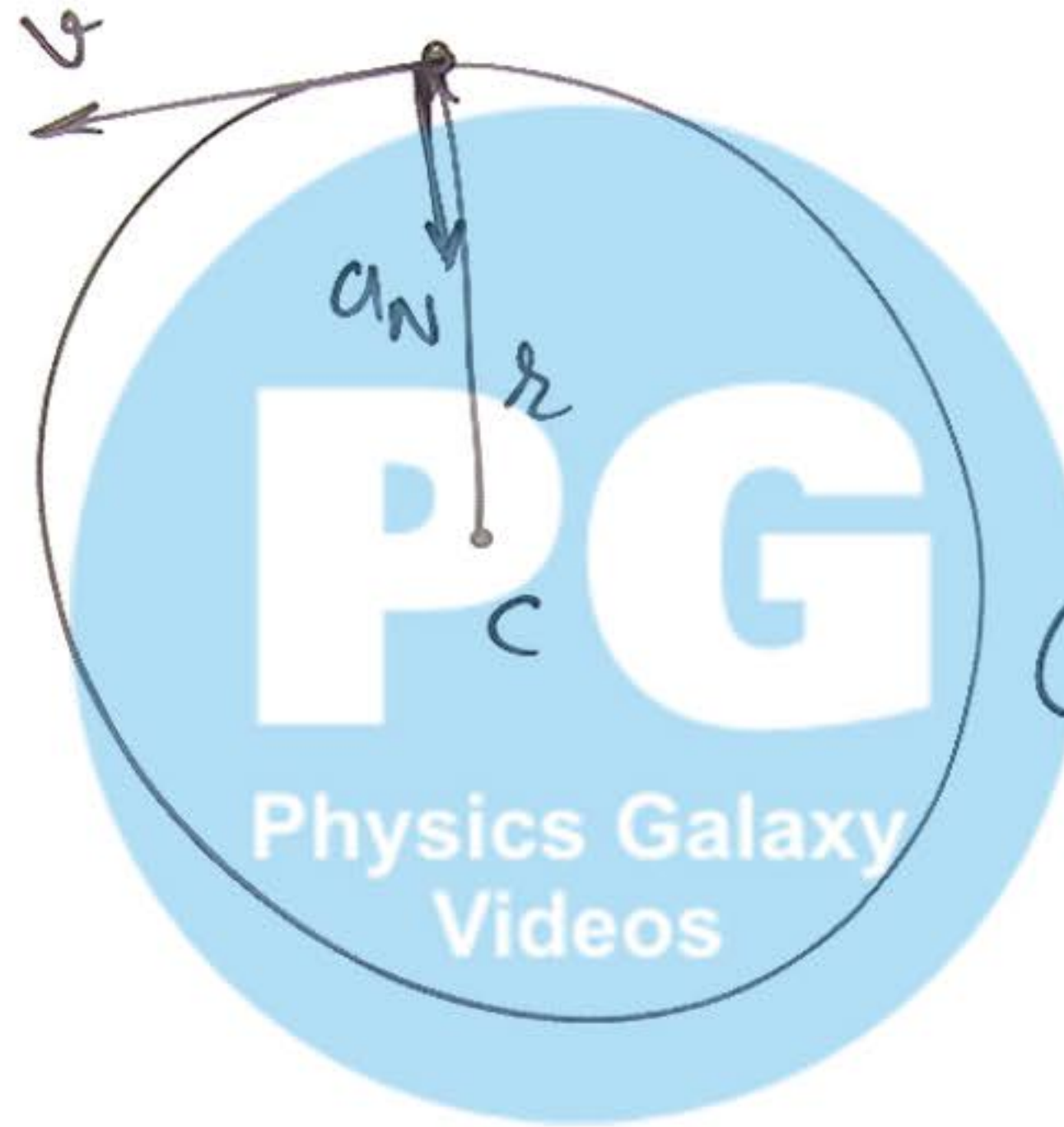
$$\omega = \frac{v}{r}$$

$$|\vec{v}| = \omega r$$

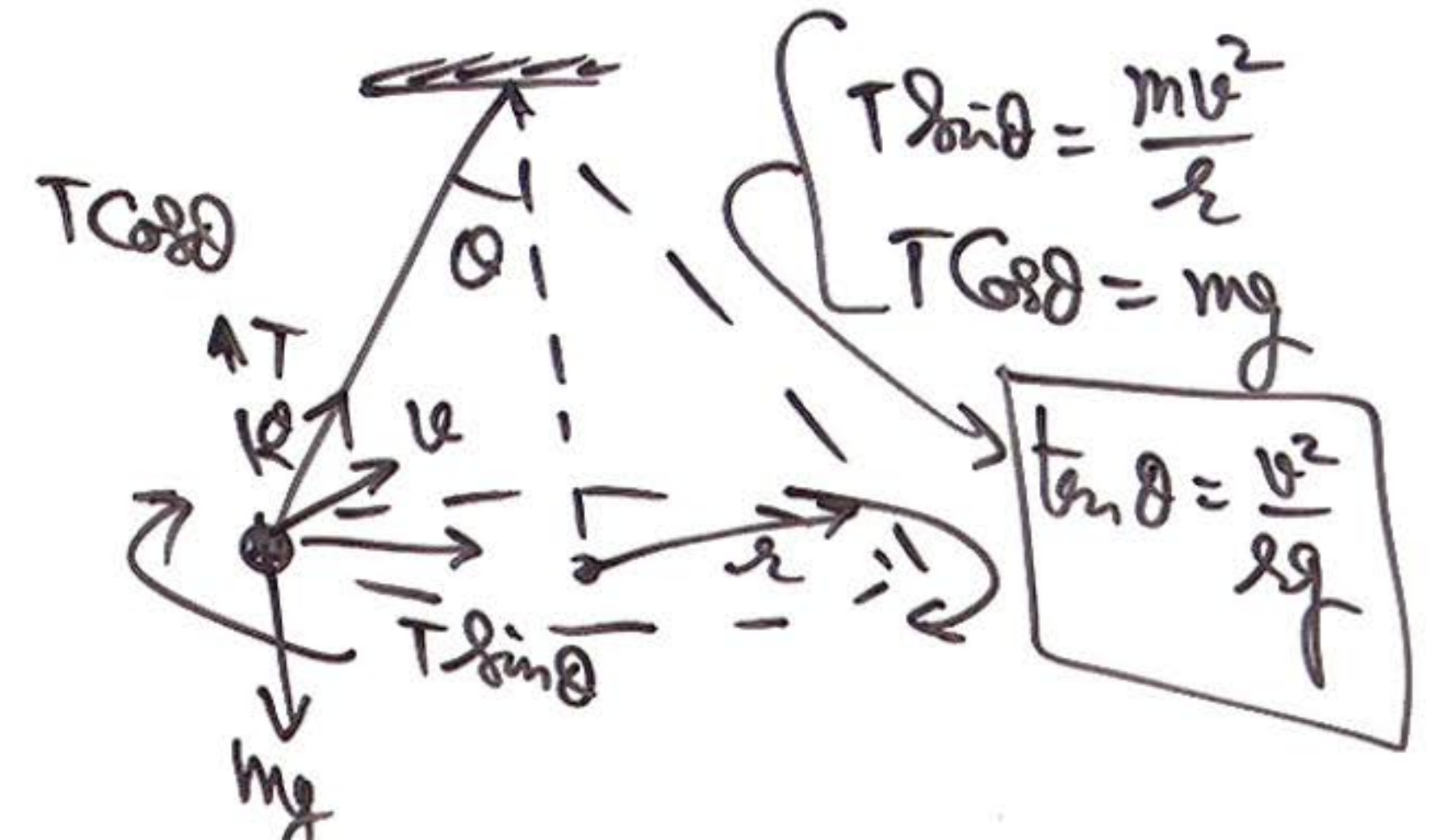
$$\underline{a_t = r\alpha} \left[\begin{array}{l} a_t = 0 \checkmark \\ \alpha = 0 \end{array} \right]$$

$$a_N = \frac{v^2}{r}$$

$$F = ma_N = \frac{mv^2}{r} \text{ [Centripetal force]}$$



Conical pendulum.



④ NUCM (Constant α):

for rectilinear motion for constant accⁿ, we use

for NUCM ($\alpha = \text{const}$), we use

$[\text{rad/s}^2]$

$$\omega = \omega_0 + \alpha t$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\Delta\theta_n = \omega_0 + \frac{\alpha}{2}(2n-1)$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

$$s = ut + \frac{1}{2}at^2$$

$$\Delta s_n = u + \frac{a}{2}(2n-1)$$

⑤ NUCM (Variable α):
 "use of Calculus is reqd"

$$\alpha = \frac{d\omega}{dt} = \omega \frac{d\omega}{d\theta}$$

IL: $\theta = 5t^3 + 2t^2 + 7 \text{ rad}$
 $\omega = 15t^2 + 4t \text{ rad/s}$
 $\alpha = 30t + 4 \text{ rad/s}^2$

IL: If in a C.M. $\alpha = 5\theta + 2 \text{ rad/s}^2$. "IC"

initial Condition

Sol: $\int_0^{\omega_f} \omega d\omega = \int_0^{\theta} (5\theta + 2) d\theta$

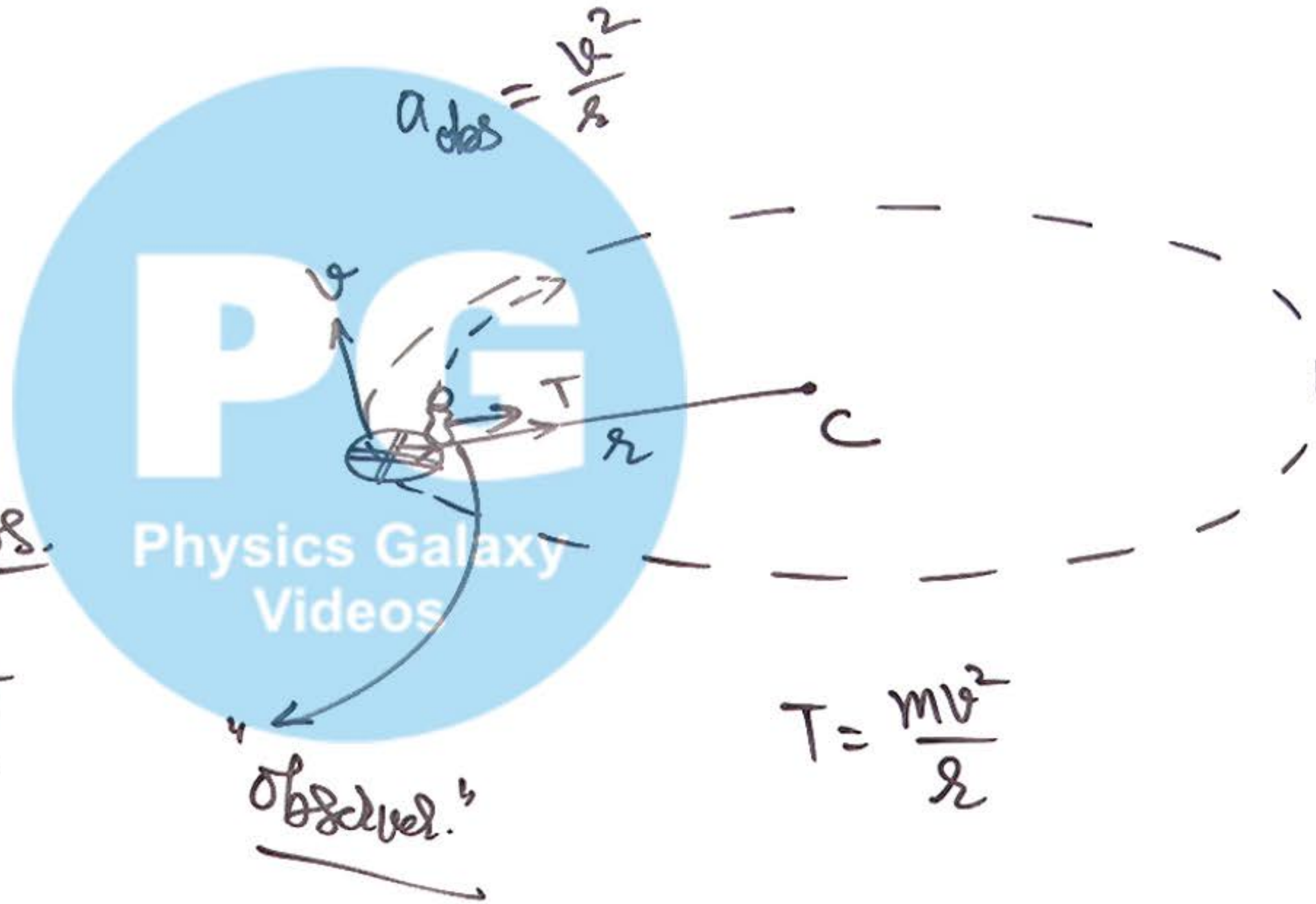
$$\Rightarrow \frac{\omega_f^2}{2} = \frac{5\theta^2}{2} + 2\theta \Rightarrow \omega = \sqrt{5\theta^2 + 4\theta} \text{ rad/s}$$

⑥ Centrifugal force:

"Observer $\rightarrow$ imp"

FBD of body
w.r. to given obs.

$m\left(\frac{v^2}{r}\right) = F$ (C.f. force)



⑦ C.M. of a Body inside a vertical Drum:

Normal force $N = \frac{mv^2}{R}$

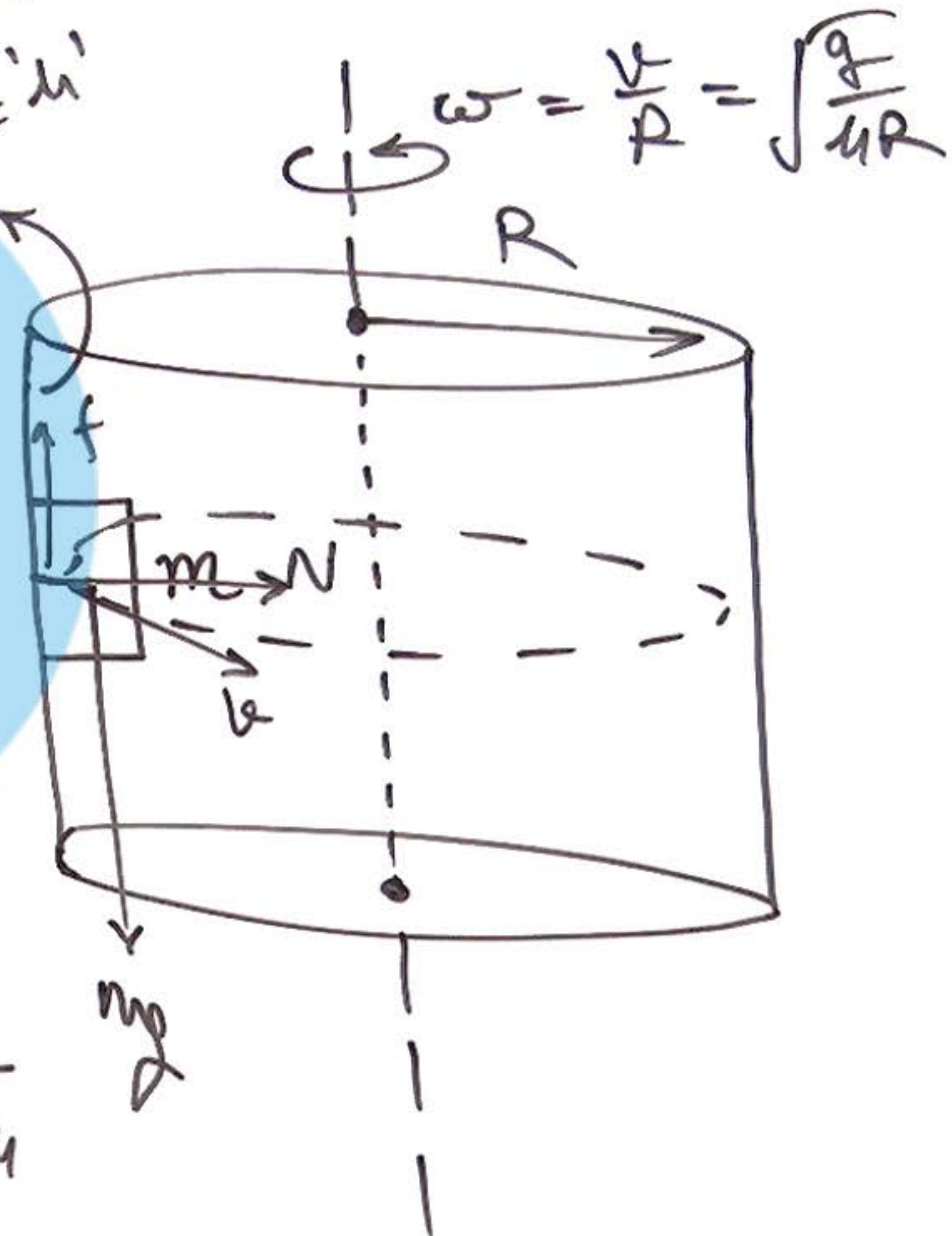
to keep body in UCM

$$4N \geq mg$$

$$4 \frac{mv^2}{R} \geq mg \rightarrow v \geq \sqrt{Rg/4}$$

friction = ' μ '

$$v = R\omega$$



⑧ Vertical C.M. of a Pendulum Bob:

Diagram illustrating the vertical circular motion of a pendulum bob of mass m and length l .

The bob is shown at two positions: at an initial angle α with velocity u , and at a final angle ϕ with velocity v . The vertical height difference between the two positions is $l(1 - \cos \theta)$.

Forces acting on the bob at angle θ are tension T and weight mg . The weight is resolved into components $mg \sin \theta$ (tangential) and $mg \cos \theta$ (radial).

Accelerations are given as:

$$a_{tp} = -g \sin \theta$$

$$a_{np} = \frac{T - mg \cos \theta}{m} = \frac{v^2}{l}$$

Using the Work-Energy Theorem:

$$\frac{1}{2} m u^2 - m g l (1 - \cos \theta) = \frac{1}{2} m v^2$$

$$v = \sqrt{u^2 - 2gl(1 - \cos \theta)}$$

Tension T is given by:

$$T = \frac{mv^2}{l} + mg \cos \theta$$

Special cases:

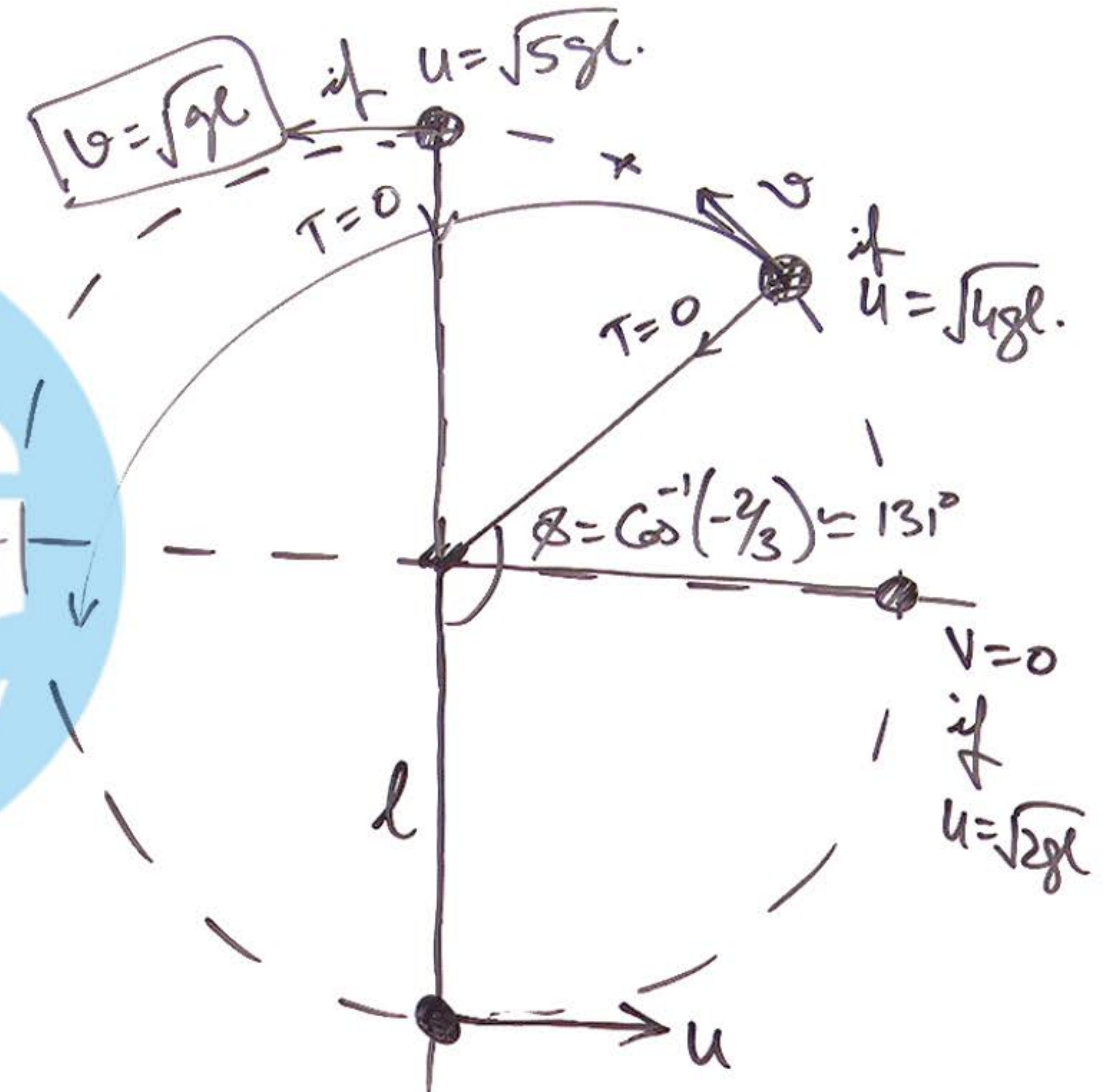
- If at $\theta = \alpha$, $v = 0$:

$$\cos \alpha = \frac{2gl - u^2}{2gl}$$
 angular amp of osc.
- If at $\theta = \phi$, $T = 0$:

$$\Rightarrow \cos \phi = \frac{2gl - u^2}{3gl}$$

9) Cases of Projection of Pendulum:

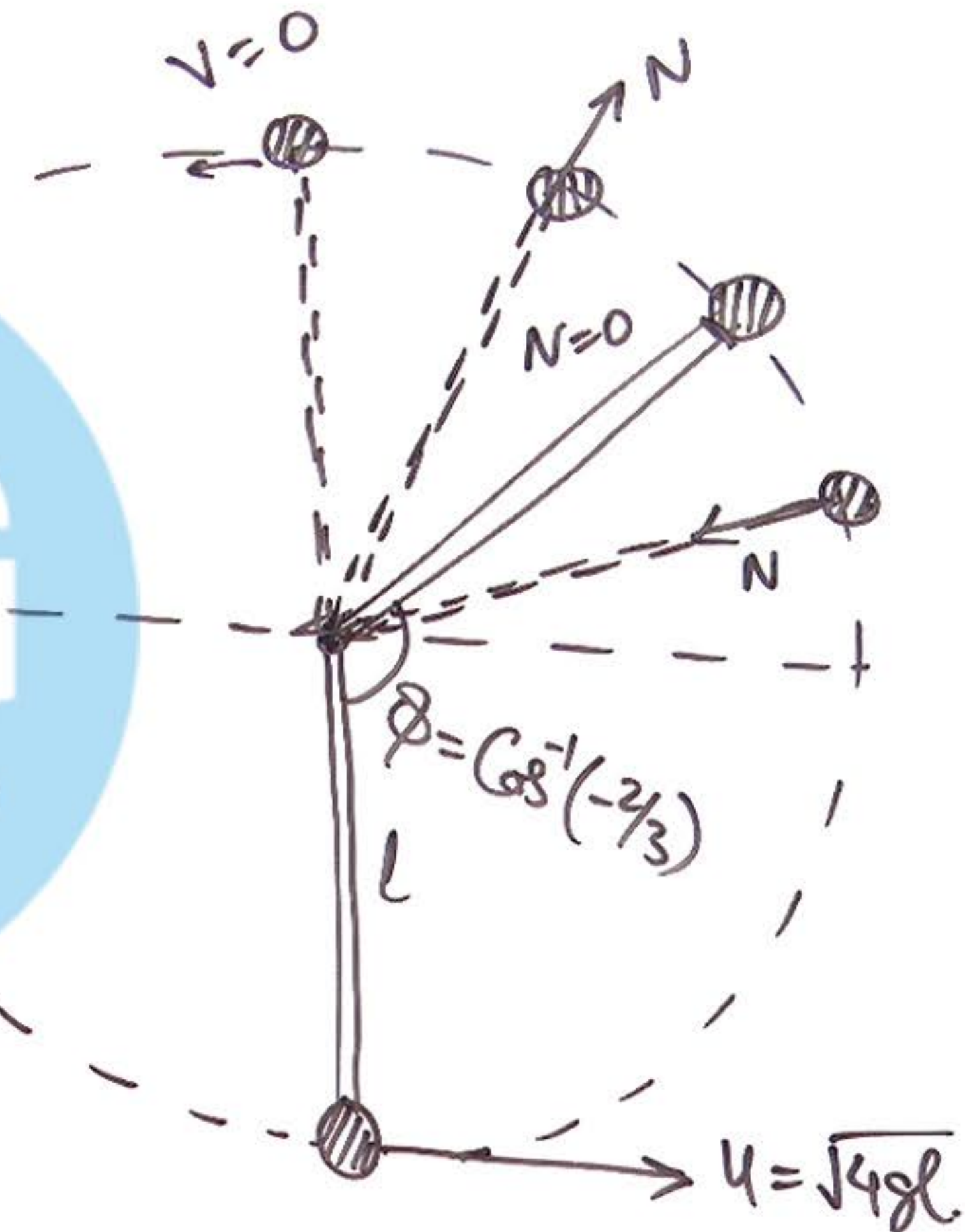
- Case-I: $u = \sqrt{2gl} \Rightarrow \alpha = \cos^{-1} \frac{2gl - u^2}{2gl} = 90^\circ$
- Case-II: $u = \sqrt{4gl} \Rightarrow \alpha = \cos^{-1}(-1) = 180^\circ$
- Case-III: $\sqrt{2gl} < u < \sqrt{4gl} \Rightarrow \beta < \cos^{-1}(-\frac{2}{3})$
- Case IV: $u = \sqrt{5gl} \Rightarrow \beta = \cos^{-1}(-1) = 180^\circ$



⑩ VC M of a light rod Pendulum:

If $u = \sqrt{4gl}$ (Min reqd vel of proj to complete the C.M)

$$\phi = \cos^{-1}\left(\frac{2gl - u^2}{3gl}\right) = \cos^{-1}\left(-\frac{2}{3}\right)$$



⑪ VCM inside a Circular Track:

$$u = \sqrt{5gR} \quad \left[\begin{array}{l} \text{min req speed of projection} \\ \text{to complete the C.M.} \end{array} \right]$$

if $u = \sqrt{4gR}$

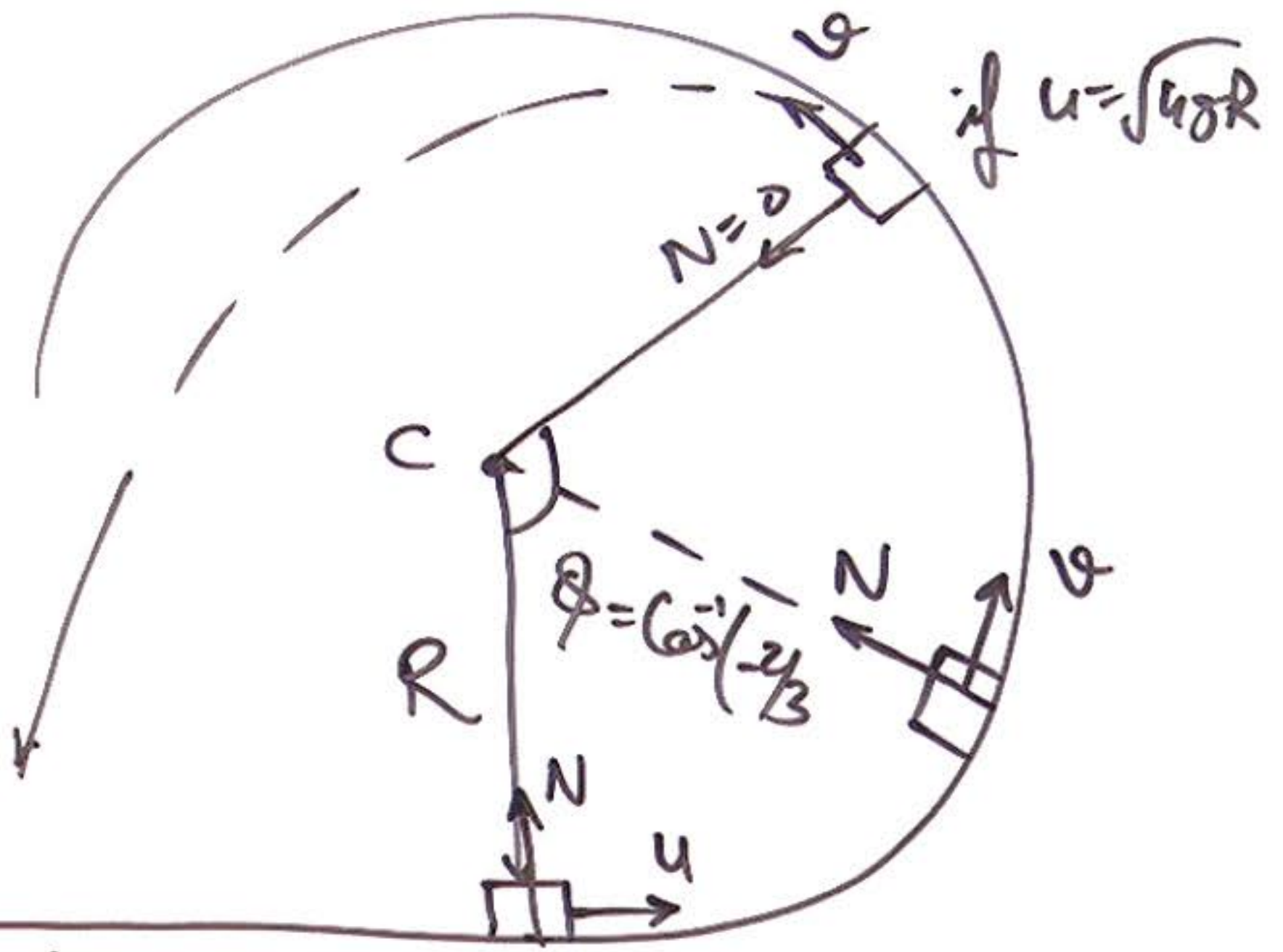
at $\theta = \phi = \cos^{-1}\left(-\frac{2}{3}\right) \approx 131^\circ$

Smooth

Back

$$\alpha = \cos^{-1}\left(\frac{2gR - u^2}{2gR}\right) \quad \left| \quad v = \sqrt{u^2 - 2gR(1 - \cos\theta)} \right.$$

$$\beta = \cos^{-1}\left(\frac{2gR - u^2}{3gR}\right) \quad \left| \quad N = \frac{mu^2}{R} - 2mg + 3mg\cos\theta \right.$$

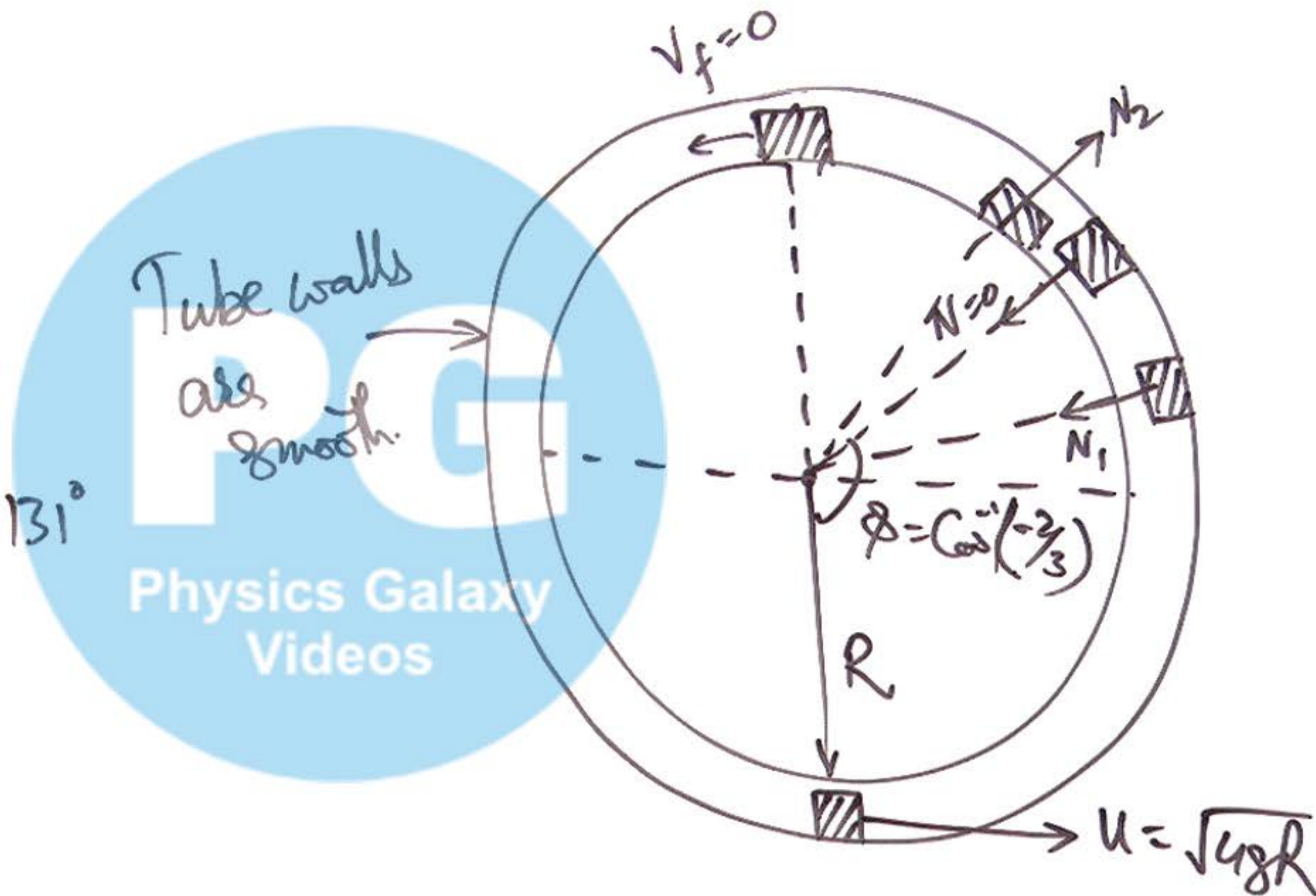


⑫ VCM inside a circular Tube :

if $u = \sqrt{4gR}$

$$\phi = \cos^{-1}(-2/3) \approx 131^\circ$$

$N=0$



⑬ VCM outside a Circular Track:

$$v = \sqrt{u^2 + 2gR(1 - \cos\theta)}$$

$$mg \cos\theta - N = \frac{mv^2}{R}$$

$$N = 3mg \cos\theta - 2mg - \frac{mu^2}{R}$$

$$N=0 \Rightarrow \theta = \cos^{-1}\left(\frac{2gR + u^2}{3gR}\right)$$

$$\text{if } u=0 \Rightarrow \theta = \cos^{-1}\left(\frac{2}{3}\right)$$

